Planning & Design Document

OPSC6312

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Introduction

This paper functions as the thorough planning and design guide for the creation of "PantryChef," a smartphone app created to address the frequently asked question, "What can I prepare with what I have?" This paper describes the specific requirements, user interface designs, system architecture, and project plan required to make the PantryChef program a reality, following the research phase in which previous recipe applications were examined. To ensure that all project stakeholders have a common knowledge of the project's scope and execution strategy, this document serves as a clear and practical guide for the development team.

App Overview

- **App Name**: PantryChef
- **Slogan:** Cook Smart, Waste Less.
- **Logo:**

- **Description**: An inventive Android software called PantryChef enables users to streamline meal planning and cut down on food waste. Users can enter ingredients that they currently have in their refrigerator or pantry. To locate delectable recipes that make use of those ingredients, the app subsequently establishes a connection with a rich recipe API. Among the salient innovative aspects are:

- **Multi-Modal Ingredient Input:** In addition to manually entering ingredients, users can add items fast by using image recognition or scanning product barcodes.
- **Smart Pantry Management:** The app keeps track of goods that are input and recommends recipes that make use of products that are about to expire.
- **Integrated Shopping List**: Simplify the grocery shopping experience by automatically creating a shopping list for any products that are lacking from recipes.
- **Customized Meal Planning**: By learning user preferences, the app may make weekly meal plan recommendations based on the user's dietary requirements and pantry contents.

Detailed Requirements List

Minimum Requirements:

- **User Authentication:**

  A strong, encrypted password and an email address must be sufficient for users to create a new account. Users must be able to enter their login information using Firebase Authentication. Use Google Sign-In (SSO) as a backup login option for the Final POE.

- **User Settings**:

  A settings panel that lets users modify dietary preferences (vegetarian, gluten-free, etc.), customize account information, establish notification preferences, and erase their local pantry cache must be included in the app.

- **REST API Connection:**

  The Spoonacular API will be accessed by the app (spoonacular, 2025). A secure proxy endpoint will be created using Firebase Cloud Functions to safely manage API keys and possibly add custom logic. This Firebase Function will be called by the application, and it will then call the Spoonacular API.

- **Offline Mode with Sync:**

  The offline persistence feature of Firebase Firestore will be utilized by the application. Offline, users can view and edit their favourites and pantries. When a connection is reestablished, all modifications will be automatically synchronized with the Firestore database (Firebase, 2025).

- **Real-time Notification:**

  Make use of Firebase Cloud Messaging (FCM) to implement push notifications. Notifications will remind users of scheduled meals or offer new recipes based on ingredients that are about to expire.

- **Multi-Language Support:**

  The app will support English and isiZulu as the second South African language. All UI strings will be externalised into resource files for easy translation.

Innovative Features:

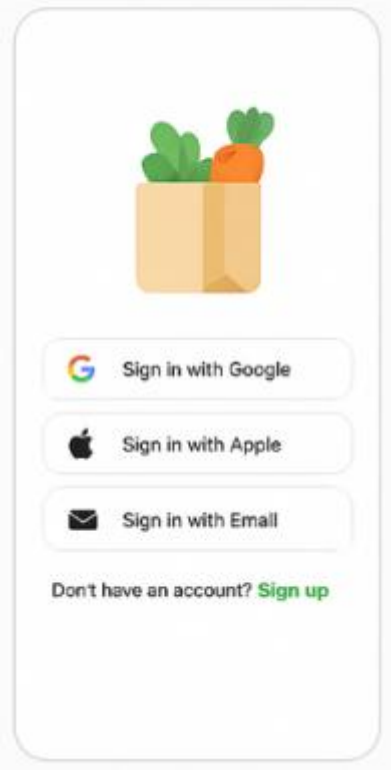
1. **Ingredient-Based Recipe Search**
   o Feature: Users can input the ingredients they have at home to find suitable recipes (SuperCook, 2025).
   o Motivation: Reduces food waste and helps users make meals without needing extra ingredients (SuperCook, 2025).
2. **Limit Missing Ingredients**
   o Feature: Users can set a maximum number of missing ingredients for recipe suggestions (SuperCook, 2025).
   o Motivation: Provides flexibility for users who only want to shop for a few extra items and make the recipes mostly from their pantry. Improves usability and convenience (SuperCook, 2025).
3. **Personalized Recipe Recommendations**
   o Feature: Suggest recipes based on dietary restrictions, allergies, and taste preferences (Yummly Recipes, 2025).
   o Motivation: Ensures the app caters to individual user needs, increasing engagement and satisfaction. It also promotes healthy and tailored cooking (Yummly Recipes, 2025).
4. **Step-by-Step Cooking Guidance with Nutrition Info**
   o Feature: Detailed instructions, timers, and nutrition breakdown for each recipe (Yummly Recipes, 2025).
   o Motivation: Helps users cook confidently, learn new recipes, and track their dietary intake (Yummly Recipes, 2025). Supports both novice and experienced cooks (Yummly Recipes, 2025).
5. **Pantry Management & Expiry Tracking**
   o Feature: When users store pantry items, they'll receive alerts for soon-to-expire ingredients (KitchenPal, 2022).
   o Motivation: Promotes efficient usage of ingredients, helps reduces food waste, and ensures that the user can plan meals effectively (KitchenPal, 2022).
6. **Suggestions Based on Available Ingredients and Expiry Dates**
   o Feature: Recommends recipes that use ingredients that are about to expire or are int the pantry (SuperCook, 2025), (Yummly Recipes, 2025), (KitchenPal, 2022).
   o Motivation: This encourages sustainable cooking habits and ensures users make the most of their ingredients (SuperCook, 2025), (Yummly Recipes, 2025), (KitchenPal, 2022).
7. **Shopping List Generator**
   o Feature: Automatically creates a shopping list for missing ingredients or weekly meal plans (Yummly Recipes, 2025), (KitchenPal, 2022).
   o Motivation: Simplifies grocery shopping by reducing forgotten items and

integrating with the recipe suggestions for a complete cooking experience (KitchenPal, 2022), (Yummly Recipes, 2025).

8. **Meal Planning & Calendar Sync**
   o Feature: Schedule meals and sync with Google Calendar (Yummly Recipes, 2025).
   o Motivation: Helps users to organize their meals and therefore helps to increase the consistency of cooking homemade meals encouraging healthier eating habits (Yummly Recipes, 2025).
9. **Favourites & User Profiles**
   o Feature: Save favourite recipes, track cooking history, and set dietary preferences (SuperCook, 2025), (Yummly Recipes, 2025).
   o Motivation: Increases user retention by creating a personalized experience and encourages repeated app usage (SuperCook, 2025), (Yummly Recipes, 2025).

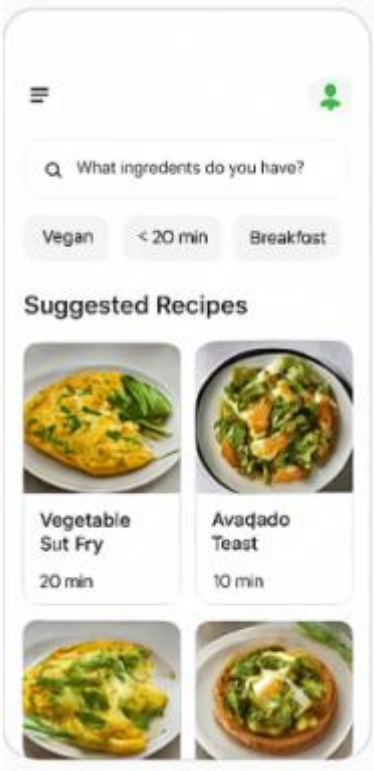
User Interface Design

Sign-Up Screen



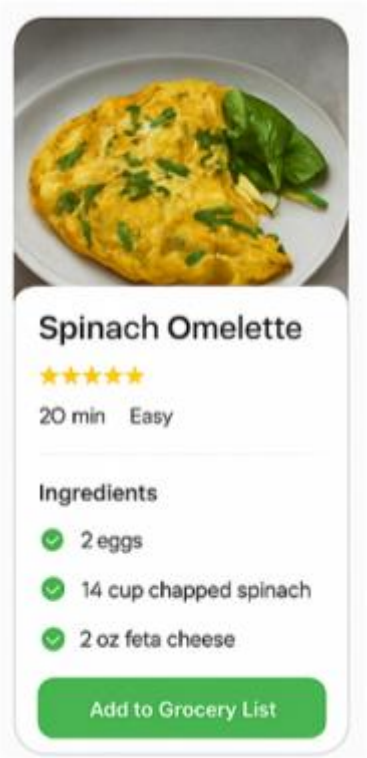
A straightforward form with buttons for "Login," "Sign-Up," and "Continue with Google" (for POE) and email and password boxes.

Recipe Results Screen



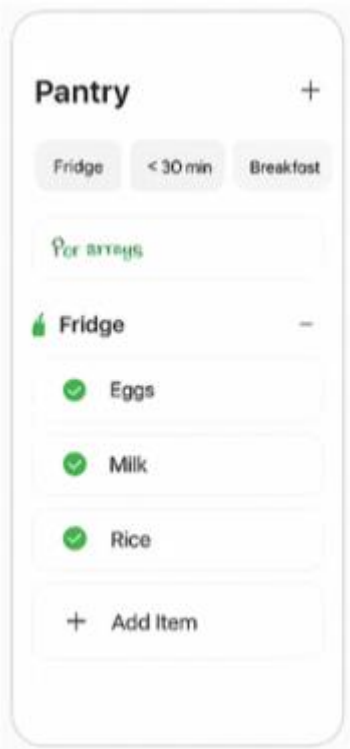
A recipe card grid with an image, title, and important information (number of missing items, ready in minutes). At the top are buttons for filters.

Recipe Detail Screen



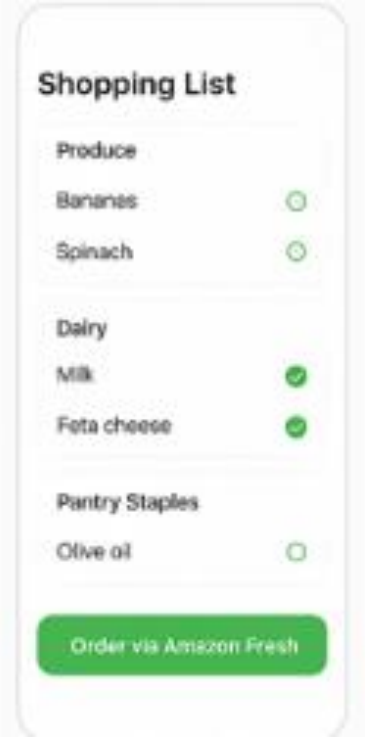
Complete recipe directions, an ingredient list, and button to "Add to Grocery List" is included in this detailed view.

Pantry Dashboard



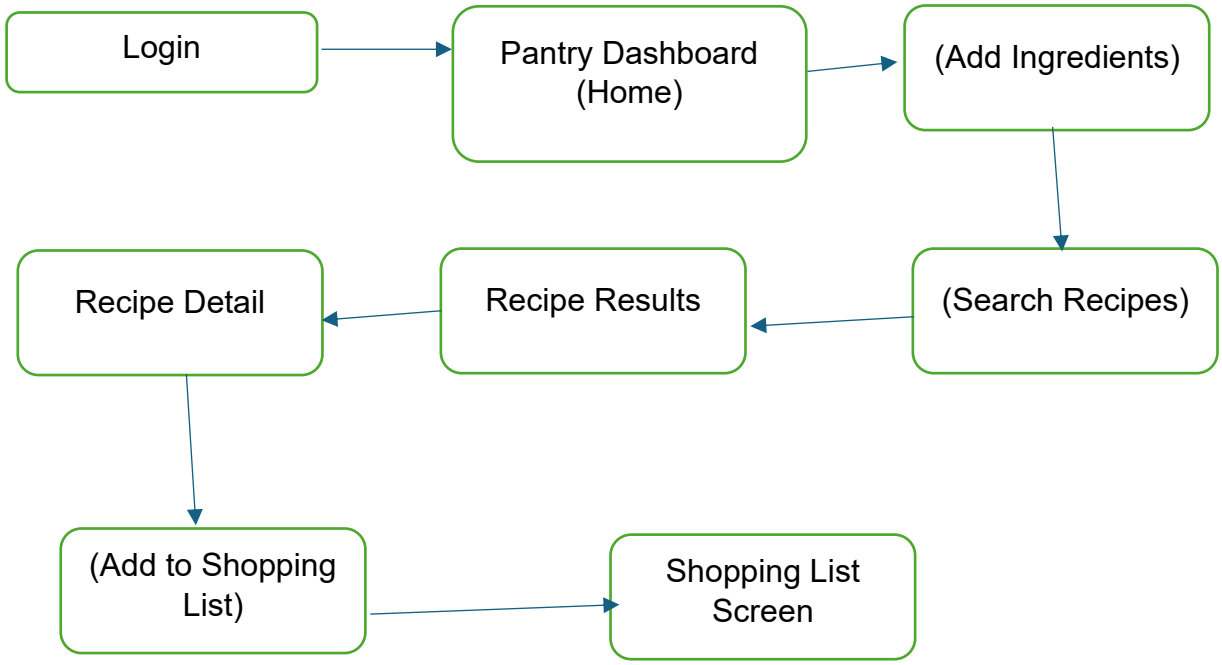
A search bar, a scrollable list of the products in the pantry, and a sizable "Add Ingredient" button are all present on the home screen. Other parts are accessible via a navigation bar at the top.

Shopping List Screen



For the produced and user-added shopping items, a straightforward checklist interface.

Navigation Flow Diagram:



API and System Design

Backend Services

The backend will be built on Google Firebase:

- **Firebase Authentication:** This will handle user registration, login and SSO (Firebase, 2025)
- **Cloud Firestore Database:** A NoSQL database will be used to store all user data, such as, dietary needs, list of pantry items, saved recipes and shopping list items (Firebase, 2025)
- **Firebase Cloud Functions:**
    - Conceal the API key from the client by serving as a secure proxy for the Spoonacular API (Firebase, 2025).
    - Might use Firestore triggers to send push messages using FCM (e.g., recommending recipes for expiring products).
- Firebase Cloud Messaging (FCM), will handle push notifications (Firebase, 2025)

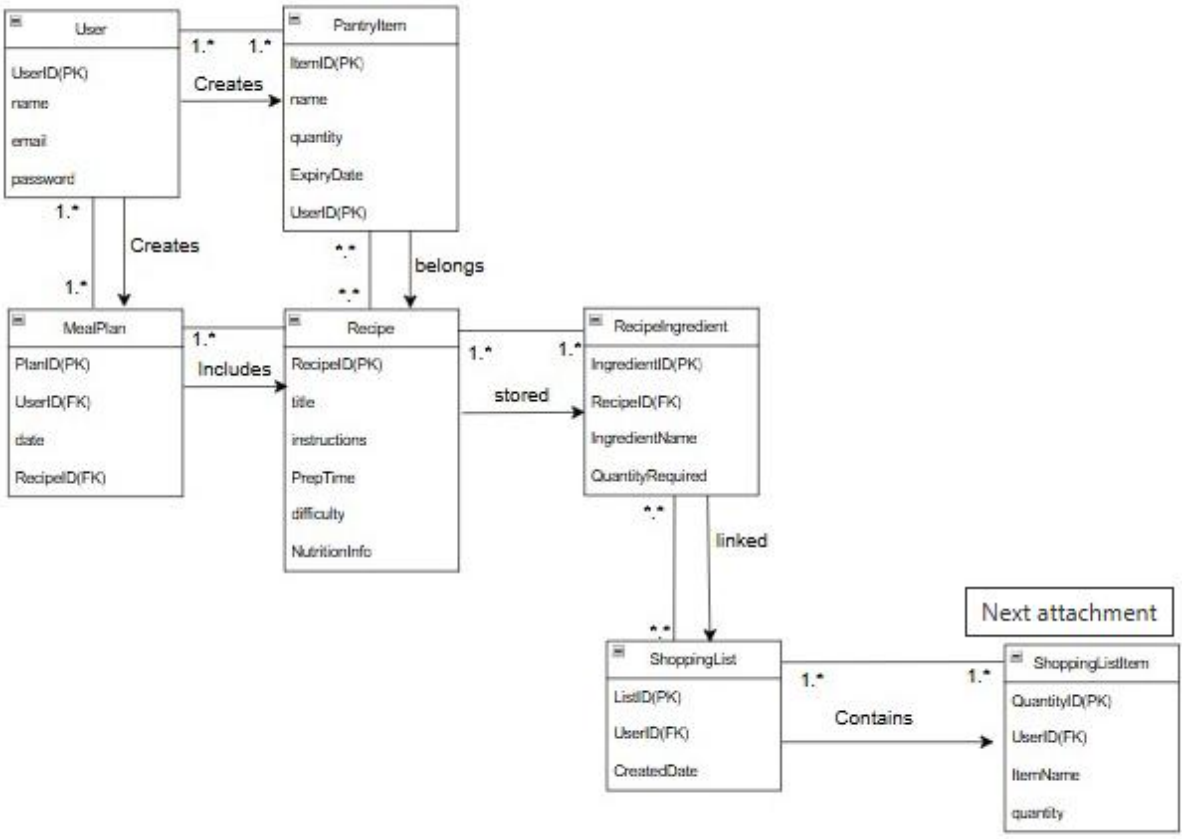
Data Flow

1. The app uses Firebase Auth to authenticate users.
2. Using offline sync, the app reads and publishes user data straight to Cloud Firestore.
3. The application submits the user's ingredient list to a Firebase Function in order to search for recipes.
4. After verifying the request and contacting the Spoonacular API, the Firebase Function evaluates the result and returns the recipe data to the application.

Hosting

Google Cloud hosts the Firebase functions automatically, therefore there is no need for a separate hosting configuration.

UML Diagram

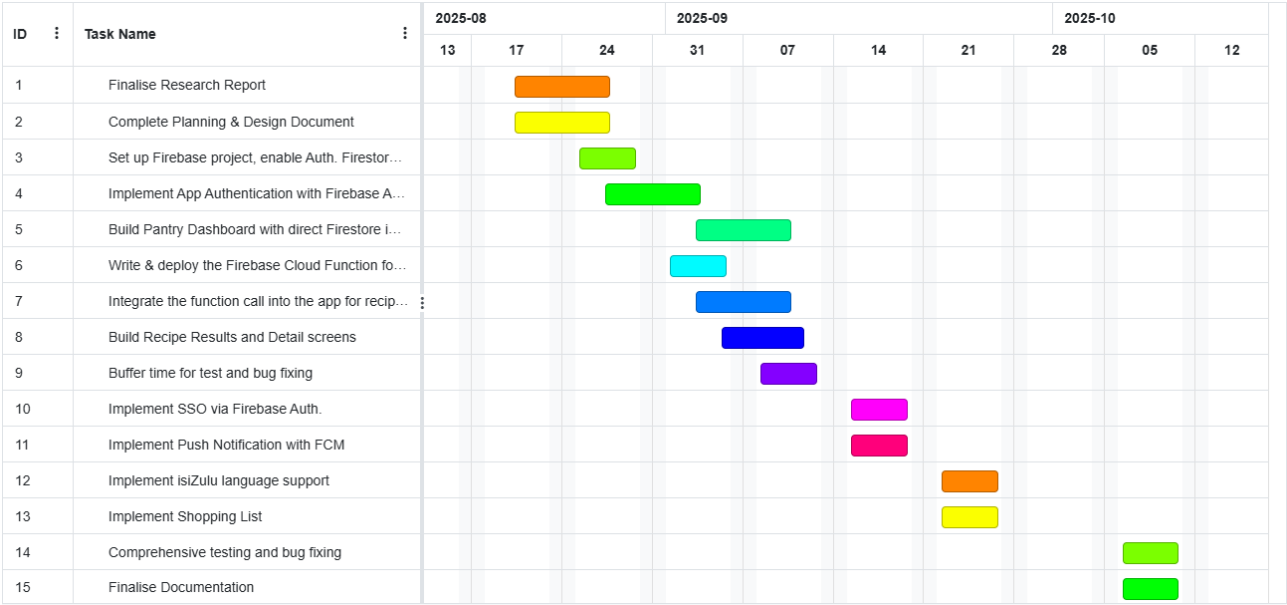


Data Model

Data will be organized into Firestore documents and collections. For security, the hierarchical structure will make use of user IDs.

- Every document ID in the Root Collection is a userId (from Firebase Auth).
  - Subcollection pantry: Items are represented by documents.
    - Fields: expiryDate (Timestamp), dateAdded (Timestamp), and name (String)
  - Favourites from the subcollection: Documents are stored recipes.
    - Fields: recipeData (Map), recipeId (Number)
  - Items are represented by documents in the subcollection shopping list.
    - Fields: isCompleted (Boolean), name (String)
  - Directly beneath the user document are the document preferences.
    - Fields: notificationsEnabled (Boolean), dietary (Array)

Project Plan



Conclusion

A thorough development plan for the PantryChef application has been laid out in this design document. A realistic project timeframe, system architecture, data management plan, user interface flow, and functional requirements are all covered in detail. Following this plan will enable the development team to create a solid, feature-rich, and easy-to-use application that satisfies all academic criteria and offers users real value while encouraging sustainable cooking practices. The build must then be carried out in accordance with this strategy, beginning with the Part 2 prototype development.

References

Firebase, 2025. *Firebase.* [Online]
Available at: https://firebase.google.com/docs/firestore?hl=en&authuser=1
[Accessed 22 August 2025].

Firebase, 2025. *Firebase.* [Online]
Available at: https://firebase.google.com/docs/cloud-messaging/concept-options
[Accessed 25 August 2025].

Firebase, 2025. *Firebase.* [Online]
Available at: https://firebase.google.com/docs/auth/
[Accessed 25 August 2025].

KitchenPal, 2022. *KitchenPal.* [Online]
Available at: https://kitchenpalapp.com/en/
[Accessed 25 August 2025].

spoonacular, 2025. *spoonacular API.* [Online]
Available at: https://spoonacular.com/food-api/docs
[Accessed 27 August 2025].

SuperCook, 2025. *SuperCook.* [Online]
Available at: https://www.supercook.com/#/desktop
[Accessed 25 August 2025].

Yummly Recipes, 2025. *Yummly Recipes.* [Online]
Available at: http://www.yummlyrecipes.com/?m=1#google_vignette
[Accessed 25 August 2025].